Linear Algebra (MT-1004)

Homework#09

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Question 1

Let

$$A = \begin{bmatrix} 5 & 4 & 2 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix}$$

be a 3×3 real matrix.

(a) Find the characteristic polynomial of the matrix A.

(b) Determine all eigenvalues of A.

(c) For each eigenvalue, find the corresponding eigenspace and its dimension (geometric multiplicity).

(d) Based on your findings, determine whether the matrix A is diagonalizable. Justify your answer